

Dec 22 Lab

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#include <iostream>

using namespace std;

void display(int a[],int n){
    for(int i=0;i<n;i++){
        cout<<a[i]<<" ";

        cout<<endl;
    }
}

void bubble_sort(int a[],int n){
    int temp;
    int isthereswap=0;
    int total_comparisons=0;
    for(int pass=1;pass<=n-1;pass++){
        for(int j=0;j<n-pass;j++){
            total_comparisons++;
            cout<<a[j]<<" "<<a[j+1]<<" ";

            if(a[j]>a[j+1]){
                //swap
                isthereswap=1;
                temp=a[j];
                a[j]=a[j+1];
                a[j+1]=temp;
            }
        }
        if(isthereswap==0)
            break;
        cout<<"output of pass "<<pass<<" is:\n";
        display(a,n);
    }
}
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}

int binary_search(int a[],int n,int key){
int left=0;
int right=n-1;
int middle;
while(left<=right){
middle=(left+right)/2;
if(a[middle]==key)
return middle;
else if(key<a[middle])
right=middle-1;
else
left=middle+1;
}
return -1;

}

int linear_search(int a[],int n,int key){

for(int i=0;i<a[i];i++){
if(a[i]==key)
return i;
}
return -1;

}

int main(int argc, char** argv) {

int a[]={10,5,7,4,6,8,1,9};
int b[]={1,2,3,4,5,6,7,8};
int choice;
char ans;
int key;

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int index;

do{
cout<<"=====MENU=====\\n" ;
cout<<"Press 1 for linear search.\\n";
cout<<"Press 2 for binary search.\\n";
cout<<"Press 3 for bubble sort.\\n";
cout<<"Your choice:";
cin>>choice;
switch(choice){
case 1:
cout<<"Enter the number you want to search:";
cin>>key;
index=linear_search(a,8,key);
if(index !=-1)
cout<<key<<" is found at index "<<index<<endl;
else
cout<<key<<" is not found.\\n";
break;

case 2:
cout<<"Enter the number you want to search:";
cin>>key;
index=binary_search(b,8,key);
if(index !=-1)
cout<<key<<" is found at index "<<index<<endl;
else
cout<<key<<" is not found.\\n";
break;

case 3:
cout<<"Before sorting using bubble sort\\n";
display(b,8);
bubble_sort(b,8);

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cout<<"After sorted using bubble sort\n";  
display(b,8);  
break;  
}  
cout<<"Do want to continue?(y/n)";  
cin>>ans;  
}while(ans!='n');  
return 0;  
}
```